framework. In relativistic quantum field theory, for example, it is straightforward to compute, in a completely Lorentz-invariant manner, the probability amplitudes and probabilities for outcomes of various experiments. But the standard treatments stop short of also describing the way in which one particular outcome or another emerges from the range of quantum possibilities—that is, what happens in the measurement process. This is a particularly important issue for entanglement, as the phenomenon hinges on the effect of what an experimenter does—the act of measuring one of the entangled particle's properties. For a more detailed discussion, see Tim Maudlin, *Quantum Non-locality and Relativity* (Oxford: Blackwell, 2002).

18. For the mathematically inclined reader, here is the quantum mechanical calculation that makes predictions in agreement with these experiments. Assume that the axes along which the detectors measure spin are vertical and 120 degrees clockwise and counterclockwise from vertical (like noon, four o'clock, and eight o'clock on two clocks, one for each detector, that are facing each other) and consider, for argument's sake, two electrons emerging back to back and heading toward these detectors in the so-called singlet state. That is the state whose total spin is zero, ensuring that if one electron is found to be in the spin-up state, the other will be in the spin-down state, about a given axis, and vice versa. (Recall that for ease in the text, I've described the correlation between the electrons as ensuring that if one is spin-up so is the other, and if one is spin-down, so is the other; in point of fact, the correlation is one in which the spins point in opposite directions. To make contact with the main text, you can always imagine that the two detectors are calibrated oppositely, so that what one calls spin-up the other calls spin-down.) A standard result from elementary quantum mechanics shows that if the angle between the axes along which our two detectors measure the electron's spins is θ, then the probability that they will measure opposite spin values is $\cos^2(\theta/2)$. Thus, if the detector axes are aligned ($\theta = 0$), they definitely measure opposite spin values (the analog of the detectors in the main text always measuring the same value when set to the same direction), and if they are set at either +120° or −120°, the probability that they measure opposite spins is $\cos^2$ (+120° or − 120°) = $1/4$. Now, if the detector axes are set randomly, $1/3$ of the time they will point in the same direction, and $2/3$ of the time they won't. Thus, over all